Relecom

AI-Powered Fraud Detection Platform

Technical Architecture & System Documentation

1. Executive Summary

Relecom is a self-learning fraud detection platform designed for telecommunications service providers. It identifies fraudulent service orders — specifically the pattern where a customer disconnects service (often due to non-payment), then reconnects under a different name at the same address to avoid the outstanding balance and capture new-customer promotions.

The system is primarily rules-based, with a machine learning layer that learns from analyst interactions over time. The more cases analysts resolve — confirming fraud or clearing false positives — the smarter the system becomes at distinguishing real fraud from legitimate orders. Every resolved case feeds back into the learning loop, continuously improving detection accuracy without manual tuning.

The platform combines a deterministic rule engine with four ML layers. Each layer analyzes orders from a different mathematical perspective, and a composite scorer blends all five outputs into a single 0-100 risk score with full explainability. No score is a black box — every risk assessment includes the individual contribution from each layer and a list of specific evidence items that drove the score.

Relecom queries and compares incoming orders against existing data in real time. It does not require bulk ingestion of a client's entire customer database — it cross-references each new order against the orders already in the system, making it lightweight to deploy and easy to integrate with existing workflows.

Key Capabilities

- **Five-layer scoring:** Rule engine, encoder embeddings for fuzzy matching, Isolation Forest for unsupervised anomaly detection, gradient boosted classifier for supervised prediction, and graph network analysis for fraud ring detection.
- **Signal-based scoring:** The rule engine only generates points from actual fraud signals (address reuse, identity matches, delinquent balances). An order with no fraud signals scores zero regardless of channel or agent.
- **Two-signal minimum:** A single fraud signal cannot push an order above Medium risk. High and Critical risk require two or more independent evidence types, reducing false positives from coincidental matches.
- **Real-time API:** REST endpoints score individual orders or batch-process entire datasets. Each scoring call returns the full evidence breakdown.
- **AI-powered investigation:** Claude (Anthropic) generates case summaries and powers an analyst chat interface on each case detail page. Falls back gracefully to rule-based summaries when the API key is not configured.
- **Self-learning feedback loop:** Resolved case outcomes (confirmed fraud vs false positive) feed back into the system to train the gradient boosted classifier. The ML layer requires approximately 2,000+ resolved cases before meaningful pattern learning begins, but the rules-based engine provides strong detection from day one. Over time, the system learns to identify emerging fraud patterns not covered by existing rules.

- **Agent tracking:** Agents are only flagged when they accumulate 3+ previously confirmed fraud orders — not simply for being third-party sales agents.

2. System Architecture

Relecom is a full-stack web application built with Next.js 14 (App Router), deployed on Vercel with auto-deploy from GitHub. The database is PostgreSQL hosted on Neon (serverless). Authentication uses NextAuth.js v5 with Google OAuth and role-based access control.

2.1 Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 14, React, Tailwind CSS	Analyst dashboard and case management UI
Backend	Next.js API Routes (serverless)	REST API, scoring pipeline, ML inference
Database	PostgreSQL (Neon serverless)	Orders, cases, users, audit log
ORM	Drizzle ORM	Type-safe queries and schema migrations
Auth	NextAuth.js v5, Google OAuth	Authentication with role-based access
AI	Anthropic Claude SDK	Case summaries and analyst chat
ML	Pure TypeScript (zero dependencies)	All ML models run in-process, no external services
Deploy	Vercel (auto-deploy from GitHub)	Push to main triggers build and deploy

2.2 Data Flow

1. **Ingestion:** Raw orders arrive via CSV upload or the /api/v1/score endpoint. Each order contains customer name, address, phone/email/payment hashes, agent ID, channel, order type (connect or disconnect), and optional fields like disconnect reason and delinquent balance. Note: SSN is not collected by Spectrum and is not used as a signal in this deployment.
2. **Normalization:** The client's existing system typically handles address validation (auto-correct, USPS-matched location IDs) before orders reach Relecom. Within the platform, addresses are uppercased and trimmed, and unit numbers (Apt, Suite, etc.) are preserved so that different units in the same building are treated as distinct addresses. The embedding layer additionally expands abbreviations (St to Street, Ave to Avenue) for fuzzy name and address comparison.
3. **Disconnect Enrichment:** For every connect order, the system searches all disconnect orders for matching normalized addresses (exact match including unit number, same zip code) and computes daysSinceDisconnect. This is the foundation of most fraud signals.

4. **Multi-Layer Scoring:** The order is scored by the rule engine, then by four ML layers (embeddings, Isolation Forest, gradient boost if trained, graph network). The composite scorer blends all outputs using configurable weights.
5. **Case Creation:** A fraud case is created with the risk score, risk band, all evidence items, financial impact estimate, and an SLA timer based on severity.
6. **Investigation:** Analysts review cases, use AI summaries and chat, and resolve as confirmed fraud, false positive, or inconclusive. Resolutions feed the learning system.

3. Scoring Layer 1: Rule Engine

The rule engine is a deterministic pattern-matching system. It evaluates each connect order against the full order pool and generates evidence items. Each evidence item has a confidence (0 to 1, how certain the signal is) and a weight (0 to 1, how important this signal type is). The score is computed as: sum of (confidence x weight) for all evidence items, multiplied by 100 and capped at 100.

> *The rule engine does not penalize orders based on sales channel. A third-party door-to-door order with no fraud signals scores zero, the same as an internal online order with no fraud signals.*

3.1 Evidence Types

The following table shows each evidence type, what triggers it, and its scoring contribution. The "Contribution" column shows the approximate points added to the score (confidence x weight x 100).

Evidence Type	Trigger Condition	Contribution
Rapid Reconnect	Connect order at an address where the same customer had a disconnect within 30 days. Urgency scales: 7 days = highest, 14 days = medium, 30 days = lower.	9-18 points
Address Disconnect Reuse	Connect order at an address that had ANY disconnect within 30 days (even a different customer). If the name is different from the prior account, confidence is higher (0.85) because this is the core slamming pattern.	18-21 points
Legacy Address Match	Address had a disconnect more than 30 days ago (e.g., 6 months ago with an unpaid balance). Only fires when there is no recent disconnect within 30 days. A weaker signal on its own, but becomes meaningful when combined with other signals like a name mismatch or identity reuse. Think of it as a flag that this address has a fraud-relevant history.	~8 points
Payment Method Reuse	Same payment hash found on a disconnected account. If at the same address, scored higher (weight 0.25, confidence 0.9). Different address is weaker (weight 0.15, confidence 0.7).	11-23 points

Phone Number Reuse	Same phone hash on a disconnected account. If under a different name, confidence is 0.85; same name is 0.65.	13-17 points
Identity Match	Two or more of 4 identity signals (phone, email, payment, equipment) match a prior account. Three+ matches = confidence 0.9.	20-27 points
Delinquency Bypass	Prior disconnect at address had a delinquent balance over $100. Strong indicator the reconnect is to avoid the debt.	~17 points
Agent Cluster	Agent has 3+ previously flagged fraud orders, OR company has 5+. Only counts orders already marked as fraud. Clean agents score zero.	~7 points
Promo Reset	Prior bill was more than 1.5x the new promo bill, within 30 days of disconnect. Indicates reconnect for promo pricing.	~10 points
Equipment Swap	Equipment serial changed at time of reconnect within 30 days of disconnect.	~11 points
Same Customer New Address	Same customer identity (via account number or 2+ identity matches) reconnecting at a different address within 30 days.	~9 points

3.2 Scoring Safeguards

- **Two-signal minimum:** If only one evidence type is present, the score is capped at 49 (Medium). This prevents a single coincidental match from triggering a High or Critical alert. At least two independent signal types are required to reach High (50+) or Critical (70+).
- **Retention channel mitigation:** Orders from the retention channel (internal win-back) are expected to show rapid reconnect and promo reset patterns. If those are the only two signals, the score is capped at 24 (Low). Retention orders only escalate if they have additional signals like delinquency bypass, name mismatch, or agent cluster.
- **No channel penalty:** The rule engine does not add points for an order being from a third-party channel. Channel type is noted in the analyst summary for context but does not contribute to the score.

3.3 Risk Bands (Rule Engine)

Band	Score Range	Response

Critical	70-100	Immediate investigation. Commission hold. 48-hour SLA.
High	50-69	Priority review. Commission hold pending decision. 48-hour SLA.
Medium	25-49	Standard review. Flag for secondary check. 7-day SLA.
Low	0-24	No action required. Auto-clear eligible.

4. Scoring Layer 2: Encoder Embedding Similarity

This layer uses character n-gram vectorization to detect name and address spoofing that exact-match rules miss. Since telecom providers typically have robust address validation (USPS-matched, auto-corrected), the primary value of this layer is catching customer name variations — the most common tactic in disconnect-reconnect fraud. It converts text strings into high-dimensional vectors and measures cosine similarity between them.

4.1 How It Works

1. **Text Normalization:** Customer names and addresses are preprocessed before comparison. Names are lowercased and trimmed. Addresses additionally expand abbreviations (St to Street, Ave to Avenue, Blvd to Boulevard, etc.) so that variant spellings produce more similar vectors.
2. **Character Trigram Extraction:** Each normalized string is broken into overlapping 3-character sequences with boundary markers. For example, the name "KINGSLEY" produces trigrams: "$$K", "$KI", "KIN", "ING", "NGS", "GSL", "SLE", "LEY", "EY$", "Y$$". This captures the shape of the name at a granular level.
3. **Vector Construction:** Each unique trigram becomes a dimension. The frequency of each trigram is the value in that dimension, creating a mathematical fingerprint of the text.
4. **Cosine Similarity:** The dot product of two vectors divided by the product of their magnitudes gives a similarity score between 0 and 1. Names like "Kingsley" and "Kings" share many trigrams and score high; completely different names like "Kingsley" and "Sarah" share almost none and score near zero.
5. **Disconnect Weighting:** Matches against names or addresses on a disconnected account are weighted heavily (up to 70 points). A name that closely resembles the prior account holder at a recently disconnected address is a strong slamming indicator. Non-disconnected matches contribute less.

4.2 What It Catches

The primary value is detecting name variations used in disconnect-reconnect fraud: "Kingsley" vs "Kings," reversed first and last names ("Martinez Robert" vs "Robert Martinez"), slight spelling differences ("Roberto" vs "Robert"), and nickname substitutions. The system also runs address similarity to catch spoofing that survives validation: transposed house numbers ("4521" vs "4512") or added unit numbers. Two orders with similar but not identical customer names at the same address are strong candidates for reconnect fraud under an alias.

5. Scoring Layer 3: Isolation Forest

The Isolation Forest is an unsupervised machine learning algorithm that detects statistical outliers without labeled training data. It works from day one with no configuration.

5.1 Algorithm

The algorithm builds an ensemble of random binary trees. Each tree randomly selects a feature and a split value, partitioning data recursively. Normal data points cluster together and require many splits to isolate. Anomalous points sit in sparse regions and get isolated quickly. The anomaly score is: s = 2^(-avgPathLength / c(n)), where scores near 1.0 are strong anomalies and scores near 0.5 are normal. A seeded pseudo-random number generator ensures reproducibility — the same input always produces the same output.

5.2 Feature Vector (10 dimensions)

Feature	What It Measures
Shared Phone Count	How many other orders in the system use the same customer phone number. A legitimate customer typically has a phone that appears on 1 order. If this phone appears on 4 others, that is statistically unusual.
Shared Email Count	Same logic as phone — how many other orders use the same customer email. Sharing an email across multiple accounts is abnormal.
Shared Equipment Count	How many other orders reference the same equipment serial number. Equipment reuse across supposedly different customers is a strong fraud indicator.
Shared Payment Count	How many other orders use the same payment method. Same card on multiple accounts under different names is a strong fraud indicator.
Shared Address Count	How many other orders exist at this exact address (including unit number). Multiple orders at the same address is unusual for residential service.
Agent Order Velocity	How many total orders this sales agent has submitted. Agents with unusually high volumes relative to peers stand out statistically.
Channel Risk Score	Numeric encoding of channel type (third-party channels get a higher value, but this is one of 10 features — it does not dominate the score)
Days Since Disconnect	Days between disconnect at this address and this connect order

Has Delinquent Balance	Binary flag (yes/no) indicating whether the prior disconnect at this address had an outstanding balance over $0
Identity Signal Overlap	Maximum number of matching identity signals (phone, email, payment, equipment) between this order and any single other order in the pool

The Isolation Forest does use channel type as one feature, but it contributes roughly 1/10th of the anomaly signal. An order is only flagged as anomalous when multiple features together place it in an unusual region of the feature space — channel alone cannot cause a high anomaly score.

6. Scoring Layer 4: Gradient Boosted Classifier

The Gradient Boosted Decision Tree (GBDT) is a supervised machine learning model that learns to predict fraud probability from resolved case outcomes. Unlike the other layers which work from day one, this layer requires labeled training data.

6.1 Training

The GBDT trains on resolved cases where analysts marked the outcome as confirmed_fraud or false_positive. The model requires approximately 2,000+ resolved cases before it can learn meaningful fraud patterns — this is by design, as smaller sample sizes would produce unreliable predictions. Until enough resolved cases exist, this layer reports "Not yet trained" and its weight is redistributed to the other four layers. Once activated, it trains a sequence of shallow decision trees where each tree corrects the errors of the previous ensemble, minimizing log-loss to produce calibrated fraud probabilities.

6.2 SageMaker Migration Path

The current implementation runs entirely in TypeScript within the application process. The feature extraction and training data format are designed to be compatible with Amazon SageMaker, so when the system scales beyond what in-process training can handle, the model can be migrated with minimal code changes.

7. Scoring Layer 5: Graph Network Analysis

The graph analysis layer detects fraud rings — coordinated groups of orders linked by shared signals that pairwise comparison would miss.

7.1 How It Works

The system builds a bipartite graph with two types of nodes: order nodes (one per order) and signal nodes (one per unique shared signal — phone hash, email hash, payment hash, normalized address, equipment ID, and agent code). An edge connects every order to each signal it possesses.

Connected component analysis via breadth-first search identifies clusters of orders linked through shared signals. Critically, two orders must share at least 2 non-agent signal nodes to be considered linked — a single shared signal is not enough. This prevents coincidental single-signal matches from inflating the graph. For example, if Order A shares a phone hash with Order B, and Order B shares a payment hash with Order C, all three are in the same connected component even though A and C share nothing directly.

Each component is scored based on its size (a cluster of 5 different orders linked together is more suspicious than a cluster of 2, because it suggests coordinated fraud across multiple fake identities), density (how interconnected the orders are — do they all share signals with each other, or just loosely through one intermediary?), and signal types (identity signals like phone or payment carry more weight than address signals, since sharing a payment method or phone number across different-named accounts is harder to explain innocently).

> *This layer catches coordinated fraud that other layers miss. If Agent PDS-007 submits 5 orders for 5 supposedly different customers, but 3 of those customers share phone numbers with each other (these are the customers' phone hashes, not the agent's), the graph layer links those orders together and flags the cluster as a potential fraud ring — likely the same person using different names, or the agent fabricating orders.*

8. Composite Scoring

The composite scorer blends all five layers into a single risk score using a weighted average.

8.1 Weight Distribution

Layer	With GBDT Trained	Without GBDT (Default)
Rule Engine	30%	35%
Embedding Similarity	20%	25%
Isolation Forest	15%	20%
Gradient Boosted Classifier	20%	0% (redistributed)
Graph Network Analysis	15%	20%

When the GBDT has not been trained, its 20% weight is redistributed proportionally across the other four layers. This means the system produces meaningful composite scores from day one using unsupervised methods, and gradually shifts toward supervised predictions as analysts resolve cases.

8.2 Composite Risk Bands

The composite scorer uses slightly different thresholds than the rule engine alone, since ML layers provide additional signal:

Band	Composite Score	SLA
Critical	80-100	48-hour SLA. Urgent priority.
High	60-79	48-hour SLA. High priority.
Medium	35-59	7-day SLA. Normal priority.
Low	0-34	7-day SLA. Low priority.

8.3 Explainability

Every composite score includes the individual score from each layer, the weight applied, whether the layer was active, a human-readable explanation of what each layer detected, and a list of ML-generated evidence items. The case detail page renders this as a visual breakdown with per-layer score bars so analysts can see exactly what drove the assessment.

9. AI-Powered Features

Relecom integrates with Anthropic Claude for intelligent investigation assistance. All AI features gracefully degrade when the ANTHROPIC_API_KEY environment variable is not configured — the system falls back to rule-based summaries generated from the evidence data.

9.1 Case Summaries

When an analyst opens a case, the system generates a summary by sending the full case context to Claude: order details, all evidence items from every scoring layer, the agent's historical performance stats (what percentage of this agent's orders are flagged), disconnect history at the address, and the composite score breakdown. Claude produces a narrative summary explaining why this case is flagged and what the analyst should investigate first.

When the API key is not configured, the system generates a deterministic summary from the evidence data — it leads with the strongest signal, adds supporting evidence, and notes the channel context. This ensures every case has a readable summary regardless of AI availability.

9.2 Analyst Chat

Each case detail page includes a conversational AI interface. The chat injects the full case context into the first message so the model can answer case-specific questions like "What other orders has this agent submitted?" or "How does this address compare to the disconnect pattern?" Conversation history is maintained throughout the session.

9.3 Feedback Learning

The feedback system analyzes resolved case outcomes to compute: precision rate (percentage of high-risk scores that were confirmed fraud), average scores by resolution type, accuracy per risk band, and false positive patterns. Based on this analysis, it recommends weight adjustments to the composite scorer to improve future accuracy.

10. Case Management

10.1 Dashboard

The main dashboard shows real-time counters for each risk band (Critical, High, Medium, Low), average risk score, and total open cases. The case queue lists all cases sortable by risk score, status, customer name, order date, agent ID, and SLA.

10.2 Case Detail

Each case page shows: order information, all evidence items grouped by source, the ML scoring breakdown with per-layer bars, the AI summary (or fallback rule-based summary), the analyst chat interface, anomaly alerts, and the full audit trail of actions taken on the case.

10.3 Resolution

Analysts resolve cases as confirmed fraud, false positive, or inconclusive. Resolutions are timestamped and logged in the audit trail. Confirmed fraud and false positive resolutions become training data for the gradient boosted classifier and feed into the feedback analytics.

10.4 Role-Based Access

Role	Permissions
Analyst	View and work cases, add comments, resolve cases
Manager	All analyst permissions plus assign cases, view team metrics, access feedback analytics
Admin	Full system access including rescoring, batch operations, user management

Authentication is handled via Google OAuth through NextAuth.js v5. An optional ALLOWED_EMAIL_DOMAIN environment variable restricts sign-in to a specific organization domain.

11. Database Schema

Table	Purpose
users	Accounts with role (analyst/manager/admin), team, login tracking
orders	Raw service orders: customer info, identity hashes, agent, channel
cases	Scored fraud cases (1:1 with orders): score, band, evidence, status, assignment, resolution
case_comments	Notes and audit entries per case
ingestion_batches	Tracks CSV upload and API batch import progress
audit_log	Immutable compliance trail of every system action

All identity fields (phone, email, payment method) are stored as one-way hashes. The system never stores raw PII in these columns — it only needs to detect when two orders share the same underlying value.

12 Roadmap

- **CRM/Salesforce Integration:** Direct API connection to the client's Salesforce (or equivalent CRM) to query real customer data, disconnect histories, and account records. The working concept is complete; production deployment requires connecting to the client's Salesforce API.
- **SageMaker Migration:** Move GBDT training to Amazon SageMaker for scalable model management and automatic retraining.

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